

Week 12 Worksheet

Electrodynamics

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Exercise 1. An infinite solenoid with a number of wire loops per unit length n is hooked up to an alternating current $I = I_0 \sin(\omega t)$. Find the electric field inside the solenoid if the radius of the solenoid is $a \ll c/\omega$.
Hint: The divergence in cylindrical coordinates is

$$\nabla \cdot \mathbf{v} = \frac{1}{s} \partial_s (s v_s) + \frac{1}{s} \partial_\phi v_\phi + \partial_z v_z.$$

Exercise 2. A toroidal coil has N tightly wound turns of wire with current $I(t) = kt$, where k is a constant. The torus has outer radius a (in the equatorial plane), and it has a circular cross section of radius $b \ll a$. Find the following fields, ignoring any components of the current which are perpendicular to the circular cross sections.

- a) The magnetic field inside and outside the solenoid.
- b) The electric field at a distance $r \gg a$ from the coil. Your answer can be of the form of the lowest multipole moment (e.g. monopole moment, dipole moment, etc.).
Hints: Maxwell's equations for Faraday fields are identical to those for *magnetostatics*, with $\mathbf{E}$ and $-\partial_t \mathbf{B}$ playing the role of $\mathbf{B}$ and $\mu_0 \mathbf{J}$, respectively.